

UTSAV BHARDWAJ

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PROFILE

Machine Learning enthusiast with expertise in algorithms, deep learning, and NLP. Skilled in Python and ML frameworks, with ongoing research work under publication in applied ML. Passionate about developing data-driven solutions and advancing AI research.

EDUCATION

Guru Gobind Singh Indraprastha University

B.Tech in Computer Science (GPA: 9.0, 6th Sem)

Dec 2022 – July 2026

New Delhi, India

SKILLS

ML & AI: PyTorch, TensorFlow, Scikit-learn, NLP, HuggingFace, OpenCV

Data Science & Analytics: NumPy, Pandas, Matplotlib, Seaborn, Power BI, Excel

Programming: Python, SQL, C++

Tools: Jupyter Notebook, Streamlit, Git

PROJECTS

HiRest – Real-Time Image Sharpening Model | PyTorch, CUDA, Adam Optimizer

june 2025

- Co-developed HiRest, a real-time image sharpening framework using knowledge distillation, published as HIRest: Real-Time Image Sharpening via Distilled HINet from Restormer with open-source implementation on GitHub.
- Designed and trained lightweight CNN (HINet) distilled from Restormer, achieving **32.81 dB PSNR**, **0.904 SSIM**, and **30+ FPS** at 720p while reducing model size to **~3.5M params** (10× smaller than teacher).
- Implemented dual-supervision training (pixel + distillation) with HIN blocks, residual decoders, CSFF, and SAM modules for computational efficiency and high-fidelity restoration.
- Validated performance using benchmark datasets and MOS evaluation (**4.1/5 vs 4.3/5** teacher), demonstrating transformer-level perceptual quality on consumer GPUs.

Spam detection using SOC | Python, DistilBert, APIs (SPF, DKIM, DMARC)

July 2025

- Co-developed an AI-powered email threat detection system leveraging NLP, Random Forest, and DistilBERT to detect phishing and spam attacks in SOC environments.
- Integrated ML classification outputs with SPF, DKIM, and DMARC authentication checks and VirusTotal threat intelligence, strengthening spoofing detection and reducing manual investigation overhead.

Blinkit Sales Analysis Dashboard | Power BI, DAX, Power Query, Excel

Nov 2025

- Built end-to-end Power BI sales analytics solution processing 8,523 transactions worth **\$1.20M**, delivering actionable insights that identified Fruits & Vegetables as top revenue driver at **\$178.12K** and revealed medium-sized outlets outperformed larger locations by **104%**.
- Designed automated DAX-based KPI framework with custom metrics matrix across 16 product categories, eliminating manual calculations and enabling stakeholders to track sales performance, average ratings, and inventory metrics in real-time.
- Executed comprehensive data transformation pipeline in Power Query to cleanse and standardize sales data, uncovering that low-fat product variants captured **65%** market share and generated **\$776.32K** versus **\$425.36K** for regular alternatives.
- Analyzed multi-year sales trends, identified 2018 as the peak year, and recommended shifting investment from large outlets to tier-2 stores to improve revenue.